

COMP 3270 – Introduction to Algorithms

Homework 5

Topic: Solving recurrence relation

1. Solve the recurrences using recursion tree with the base case as 1.

a. $T(n) = 3T(\sqrt{n}) + \theta(n^2)$

b. $T(n) = 4T\left(\frac{n}{2} + 2\right) + n$

c. $T(n) = 3T\left(\frac{n}{2}\right) + n$

d. $T(n) = T(n - 1) + T\left(\frac{n}{2}\right) + n$

e. $T(n) = 2T(n/4) + \sqrt{n}$

2. Solve using master theorem.

a. $T(n) = T\left(\frac{n}{2}\right) + \theta(1)$

b. $T(n) = 7T(n/3) + n^2 - n$

c. $T(n) = 4T(n/2) + n^2 \log n$

d. $T(n) = 2T(n/4) + \sqrt{n}$

e. $T(n) = 7T\left(\frac{n}{8}\right) + \log n$